

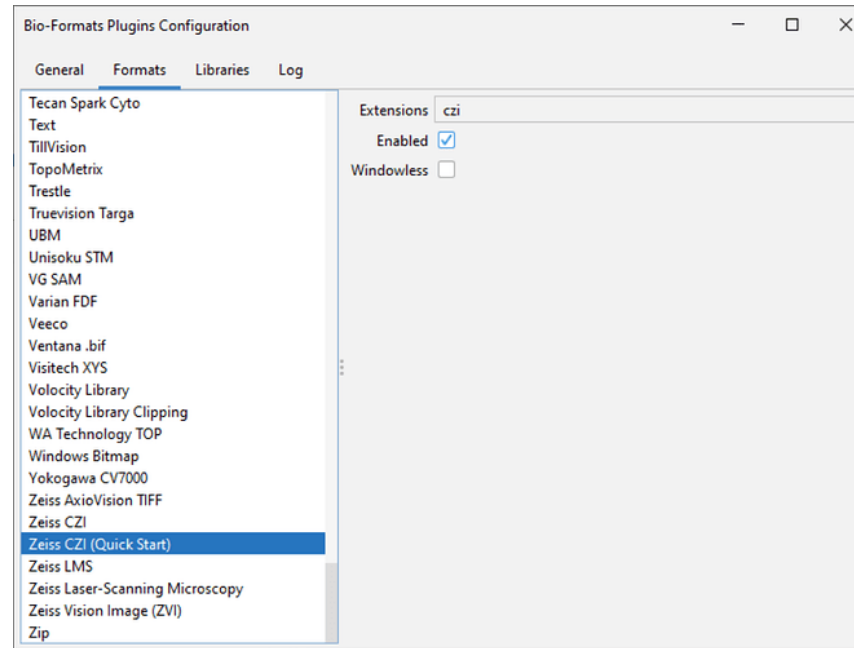
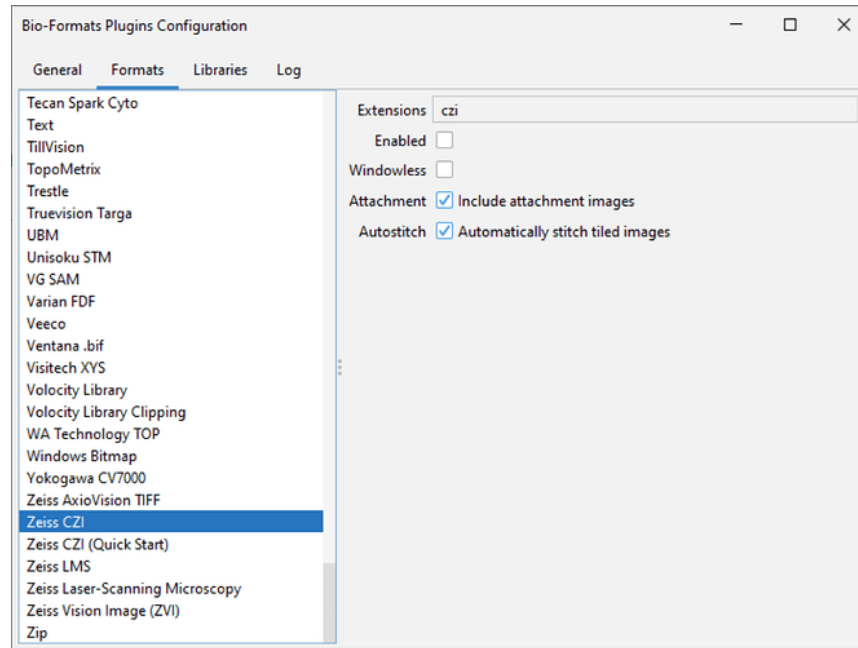
Fiji

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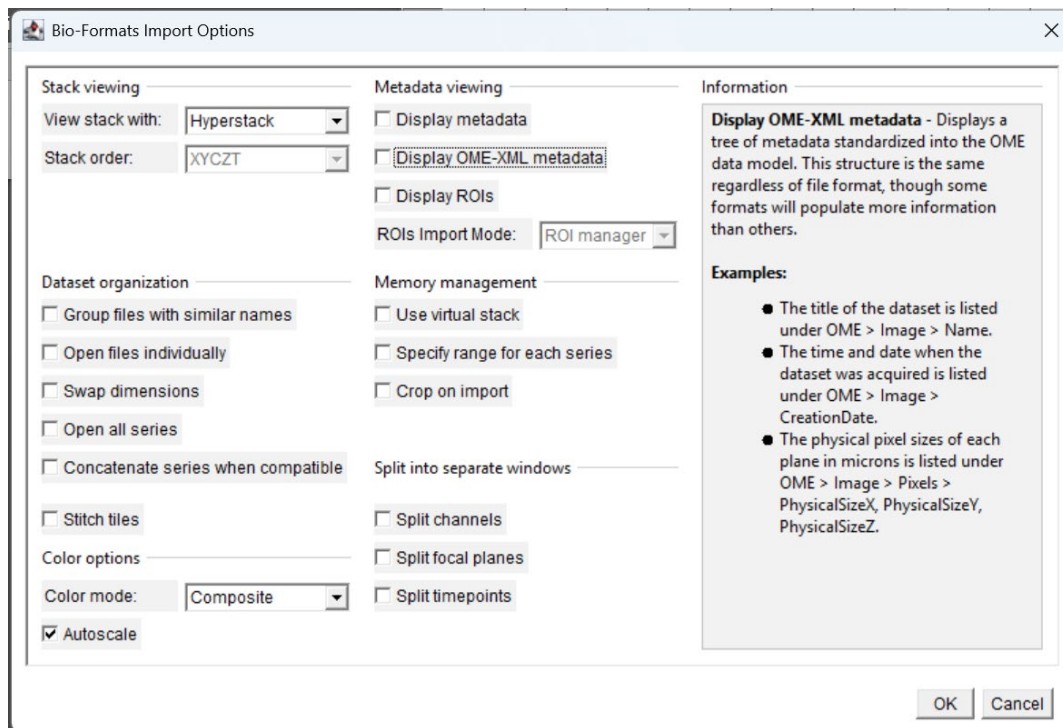
Get the high resolution image

(needed for segmentation) = convert czi file to tiff

- Open Fiji
- Help > Update... > Manage Update Sites > tick “BioFormat” & “Zeiss Quick Start Reader”
- Plugins > Bio-Formats > Bio-Formats Plugins Configuration > Formats > check it is as follows:



- Plugins > Bio-Formats Importer > select the czi files present in
/pasteur/helix/projects/CSD_hpc/data/lab/spatial/embryo_head_VisiumHD/images/
Check it is as follows (next slide)

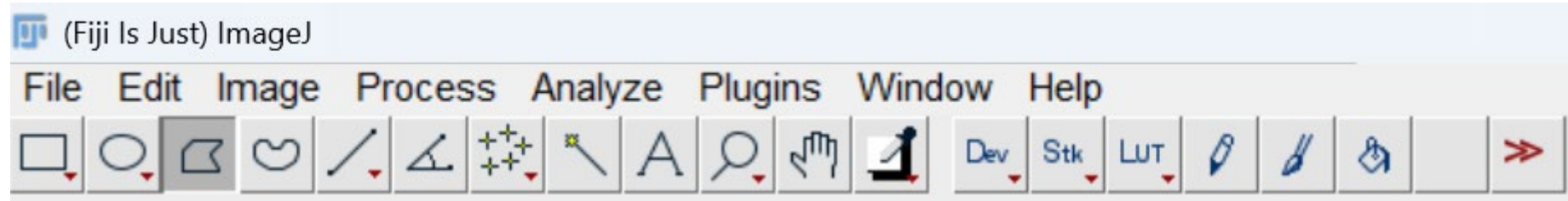


Click OK

- Select the second series then (the image having the second higher nb of pixels)
- Image > Type > RGB color
- File > Save as > tiff

Open D1_highres_RGB.tif onto Fiji for example

Select following tool:



Draw the ROI on the image:

Right click on the ROI and add to ROI Manager

Open ROI Manager through Analyze > Tools > ROI Manager

Tick "show all"

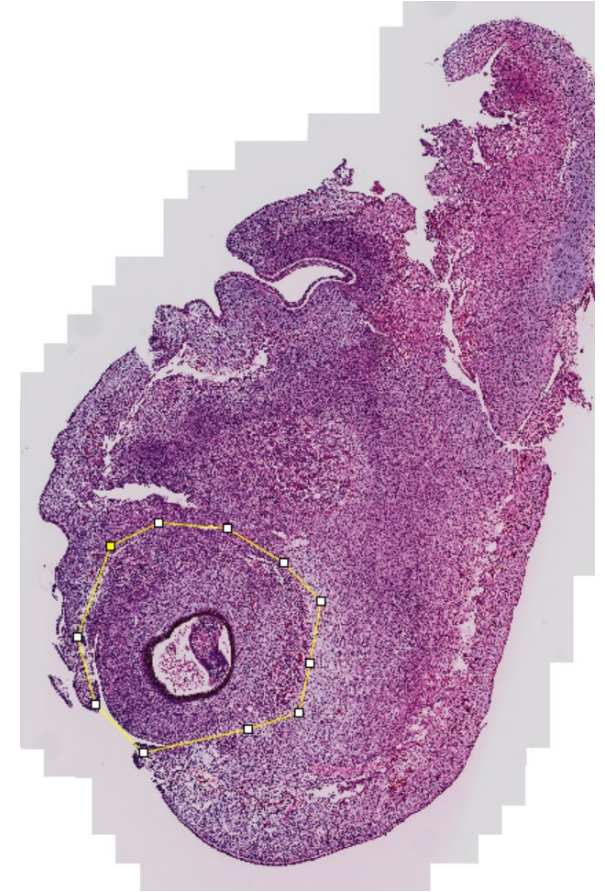
If you want the coordinates (to copy paste onto Jupyter Lab notebooks)

: click on the ROI in the ROI manager
and then Properties > List Coordinates

Or the best is also to save the ROI as a roi file directly from the ROI Manager to be able
to reload it easily later on the image to visualize the ROI.

But I've created a macro code also to simply visualize the coordinates:

See next slide



- 1) Open H&E image on Fiji: CSD_hpc/data/lab/spatial/embryo_head_VisiumHD/D1_highres_rgb.tiff
- 2) Open gene expression image generated in Markers_Visualization.ipynb and stored in fast_npz/Naomie/Shahragim_Project/visium_HD/spatial_EOM/segmentation/StarDist/embryo_head/D1/markers_Glenda/MYOG.tiff
- 3) Flip Vertically (see section “Flip Vertically Image” below) the MYOG image to match the orientation of the H&E

Note: I did not manage to overlay both images knowing they have different channels and coordinates because of distinct resolutions.

- 4) Put both images side by side and select the ROI (see “Create ROI” section above) on the H&E by looking at where MYOG is expressed on the other image
- 5) Save the ROI from the ROI Manager by renaming all ROI according to the muscle selected, and selecting all of those ROI, go to More > Save and save it as a zip in fast_npz/Naomie/Fiji to easily load them back again later
- 6) Select each ROI one after another in the ROI Manager, for each:
Go to Properties > List Coordinates > Copy the coordinates and paste them into the Notebook in bbox=Polygon([])

First make sure you loaded the image.

It's not working because setRoi does not exist. Try and see test.ijm on Desktop

Then follow:

Display a previously created ROI that was copied pasted onto Jupyter Notebook:

We have something like:

Or it was directly saved onto a csv file

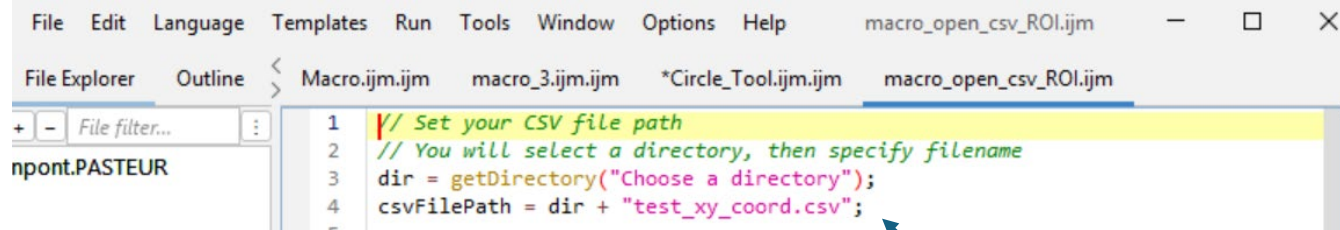
(through ROI Manager > Properties > List coordinates > Save > csv file)

Or you manually put those coordinates onto a csv file with no header

And then run the macro:

Plugins > New > Macro > Open > fast_npz/Naomie/Fiji/macro_open_csv_ROI.ijm

Then on the Macro Window:



> Run

Note: you should modify the name of csv file according to your: test_xy_coord.csv

Advice: store the csv files onto fast_npz/Naomie/Fiji/bbox_D1_EOM.csv for e.g.

```
bbox_EOM = Polygon([(3872,23574),
(4404,22270),
(7620,21334),
(11364,22630),
(12420,25246),
(12228,27814),
(11268,28678),
(8608,29718),
(6048,29910),
(5052,29446),
(4428,28798),
(4212,28198),
(3876,27046),
(3616,25782)])
```

Right click on Image > Find Commands > Flip Vertically